

PROJECT REPORT

tidepool

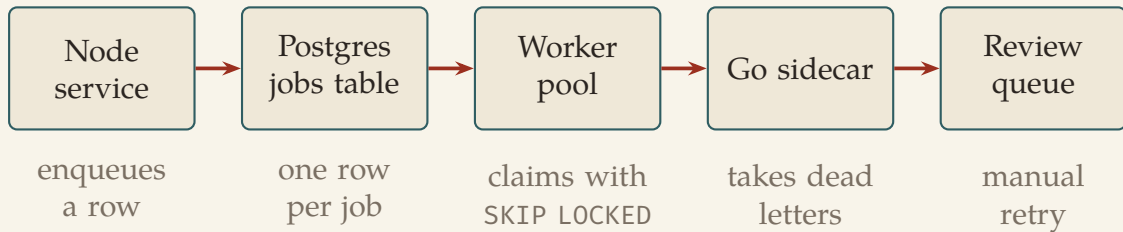
A small self-hosted job queue for Node services

Asha Rao, Leo Brandt

September 2026

tidepool

How it works



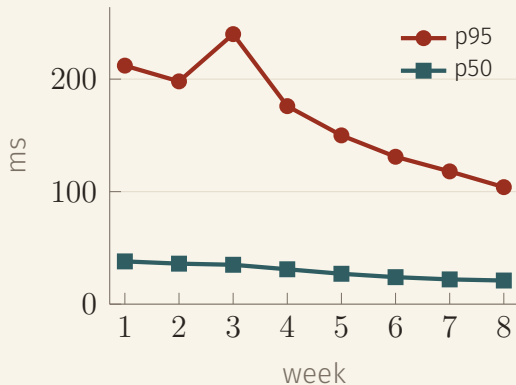
- Jobs are rows, so two workers never take the same job and no lock server is needed.
- Failed jobs back off and retry up to five times before the sidecar takes them.

Results after eight weeks

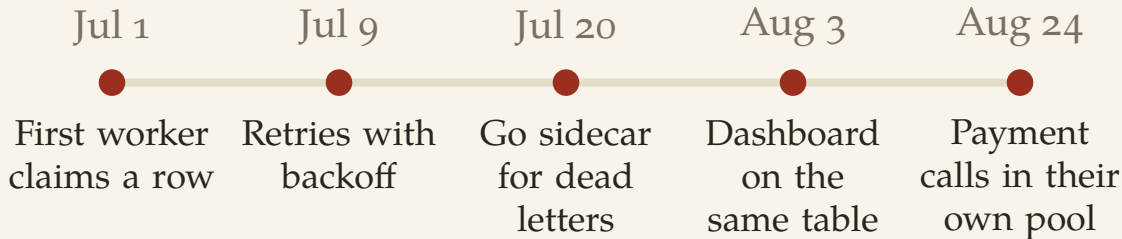
104 ms

P95 ENQUEUE TO START

Down from 212 ms in week one, after tuning the claim query and the index on run_at.



Eight weeks of work



Most failures were timeouts calling a slow payment API. Moving those calls to their own worker pool **removed about half of them.**